INCH-POUND
MIL-DTL-24643/23G
1 October 2009
SUPERSEDING
MIL-DTL-24643/23F
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +105 °C, 300 VOLTS, TYPE LSTTSU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight with Circuit Integrity

First	-	Copper conductor, coated or uncoated, 22 AWG, Class K stranding.
Second	-	Thermoset insulation, minimum average wall thickness 0.017 inch.
Third	-	Insulation jacket of polyamide, in accordance with ASTM D4066, colored (see fourth) or transparent at manufacturer's option. Minimum average wall thickness of 0.003 inch. Diameter over insulated and jacketed conductor shall be 0.076 inch minimum. Twisted pair identification code applied by Method 6.
Fourth	-	Two conductors, one white, one black, twisted together to form a pair having a left-hand lay length not greater than 2¼ inches. For type LSTTRSU-1½ only, three conductors, one white, one black, and one red twisted together to form a triad having a lay length no greater than 2¼ inches.
Fifth	-	The specified number of pairs (see table I) shall be cabled together with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with no. 1 from the center outward. Outer most layer shall be left-hand. Fillers may be used as necessary to form a firm, well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/23	Type and size	Number of pairs	Jacket thickness min. avg. (inch)	Cable overall diameter max. (inches)
-01UN	LSTTSU-1-1/2	1-1/2	0.050	0.330
-02UN	LSTTSU-3	3	0.050	0.450
-03UN	LSTTSU-5	5	0.050	0.540
-04UN	LSTTSU-10	10	0.062	0.675
-05UN	LSTTSU-15	15	0.062	0.800
-06UN	LSTTSU-20	20	0.062	0.870
-07UN	LSTTSU-30	30	0.075	1.080
-08UN	LSTTSU-40	40	0.075	1.200
-09UN	LSTTSU-50	50	0.075	1.400
-10UN	LSTTSU-60	60	0.075	1.450

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	18.60
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Abrasion resistance (scrapes, min.)	250
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Crack resistance	No damage

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero

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Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Gas flame (1 hour) (see note below)	No failure
Group C:	
Physicals (aged) air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure

NOTE: For gas flame test of LSTTSU-1½, the three connectors shall be connected to phases A, B, and C of the power supply. For test of all other sizes, connection of conductor pairs shall be in accordance with the following pattern:

a. Pair #1: One conductor to phase A, one conductor to phase B
b. Pair #2: One conductor to phase C, one conductor to phase A
c. Pair #3: One conductor to phase B, one conductor to phase C
d. Pair #4: One conductor to phase A, one conductor to phase B
e. Pair #5: One conductor to phase C, one conductor to phase A
f. Pair #6 through Pair #60: In similar sequence.

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (minus 20±2 °C)	No damage
Gas flame (3 hours) (see note 1)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure

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Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSTTSU	1½ through 10	2500
LSTTSU	15 and 20	1500
LSTTSU	30 through 60	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-027)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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